ROLE OF DIET IN ORAL HEALTH

(CALCIUM, PHOSPHATE, VITAMINS)

Learning Objectives

By the end of this presentation, students should be able to:

- ❖ Explain the relationship between diet and oral health
- ❖ Understand the biochemical basis of dental caries
- ❖ Identify cariogenic and protective foods
- ❖ Describe the role of nutrients in teeth and periodontal health
- ❖ Discuss preventive dietary strategies

Introduction

- Diet directly affects oral tissues and dental health.

- Nutrients influence:
  - Tooth development
  - Salivary composition
  - Oral microbial activity
  - Periodontal health

- Poor dietary habits increase risk of:
  - Dental caries
  - Gingivitis
  - Periodontitis
  - Enamel erosion

Oral Health and Nutrition Two-Way Relationship

- Nutrition affects oral tissues.
- Oral diseases can affect food intake and nutrition.

Examples:

- Tooth pain → reduced eating
- Vitamin deficiencies → oral lesions

Structure of Tooth and Nutritional Importance

- Enamel: highly mineralized.
- Dentin: collagen + minerals
- Cementum and bone need proper nutrition
- Important nutrients:
  - **Calcium**
  - **Phosphorus**
  - **Vitamin D**
  - **Protein**
  - **Fluoride**

Dental Caries:

- Dental caries is a multifactorial infectious disease causing demineralization of tooth structure.
- Main factors:
    - Cariogenic bacteria
    - Fermentable carbohydrates
    - Susceptible tooth

Stephan Curve and pH Drop

- After sugar intake:
  - Oral bacteria ferment carbohydrates
  - Acid production lowers plaque pH
- Repeated acid attacks cause demineralization

Biochemistry of Dental Caries

- Streptococcus mutans metabolizes sugars
- Produces lactic acid
- Acid dissolves hydroxyapatite crystals

Cariogenic Foods

Foods that Promote Caries

- Sticky sweets
- Chocolates
- Soft drinks
- Sugary snacks
- Refined carbohydrates

Protective Foods

Foods Protective Against Caries

- Cheese
- Milk
- Nuts
- Fibrous fruits and vegetables

Benefits:

- Stimulate saliva
- Neutralize acids
- Provide minerals

Remineralization

- Repair of Early Enamel Lesions
- Demineralization ⇌ Remineralization

- Factors promoting remineralization:
  - Saliva
  - Fluoride
  - Calcium-rich diet

Role of Fluoride

Benefits:

- Increases enamel resistance
- Enhances remineralization
- Inhibits bacterial metabolism
- Formation of fluorapatite

Vitamins and Oral Health

- Vitamin role in oral health has already discussed in detail.

Mineral Deficiencies and Oral Effects

Mineral	Deficiency Effect
Calcium	Poor tooth mineralization
Phosphorus	Weak enamel
Iron	Glossitis, mucosal changes
Zinc	Delayed healing

Oral cancer

- Oral cancer includes cancers of the lip, other parts of the mouth and the oropharynx and combined rank as the 13th most common cancer worldwide. (WHO)

- Tobacco, alcohol and areca nut (betel quid) use are among the leading causes of oral cancer.

- In North America and Europe, human papillomavirus infections are responsible for a growing percentage of oral cancers among young people.

Diet and Periodontal Disease

- Poor nutrition contributes to:
  - ❖ Reduced immunity
  - ❖ Delayed wound healing
  - ❖ Increased inflammation
- Risk factors:
  - ❖ Vitamin C deficiency
  - ❖ Protein deficiency
  - ❖ High sugar diet

Diet Counseling in Dentistry

- ❖ Reduce sugary snacks
- ❖ Avoid frequent sipping of soft drinks
- ❖ Eat balanced meals
- ❖ Increase water intake
- ❖ Use fluoride toothpaste

Preventive Dietary Recommendations

- Key Recommendations
  - ❖ Limit sucrose intake
  - ❖ Prefer non-cariogenic snacks
  - ❖ Consume dairy products
  - ❖ Maintain hydration
  - ❖ Eat fibrous foods
  - ❖ Avoid bedtime sugary foods

Take home message

- Oral health and nutrition are closely interconnected
- Proper dietary habits help prevent oral diseases
- Early education on nutrition is essential for dental professionals

THANK YOU